

AI BADGE



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AI Foundations 101



Ever wondered why AI confidently makes things up?

This lesson is the mental model of how a large language model actually works. Pattern matching, training data, tokens, context windows, temperature, and the reason it hallucinates.

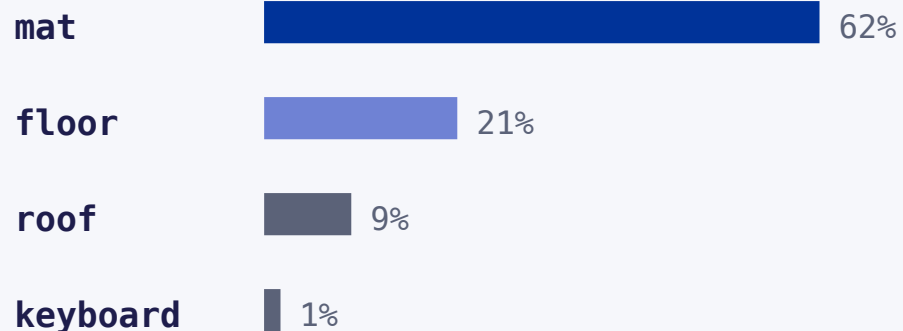
By the end, you will never be surprised by an AI mistake again.

Six ideas. One 60-second demo each. No maths, no jargon.

It predicts the next word.

Not (in its core mechanism) thinking the way you do.

The cat sat on the ____



It scores every possible next token and picks from the top. Then it does it again. One token at a time.

Why this matters

- **It is fluent, not factual**

Fluency comes from patterns in the text. Truth has to be checked separately.

- **It has no memory of you**

Each turn, it only sees what is in front of it. Nothing is recalled by default.

- **It is confident either way**

A right answer and a wrong one are produced the same way. Confidence is not a signal.

It learned the world up to a date.

After the training cutoff, it knows nothing unless you tell it or it can search.

INSIDE THE TRAINING DATA

- How language, code and concepts work
- Public facts up to the cutoff date
- Patterns across billions of documents

AFTER THE CUTOFF

- Last week's news or this morning's price
- Your private files, unless you share them
- Anything that happened after it was trained

Ask a model what day it is and it often guesses. The fix is to give it the context, or use a tool that can look things up.

AI reads text in chunks, not words.

A token is roughly four characters. About 750 words is about 1,000 tokens.

"Tokenisation is surprisingly important."

Token

isation

is

surpris

ingly

important

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One long word splits into several tokens. Common words are one token. Spaces and punctuation count too.

Cost

Most tools bill per token, in and out. Longer prompts cost more.

Length

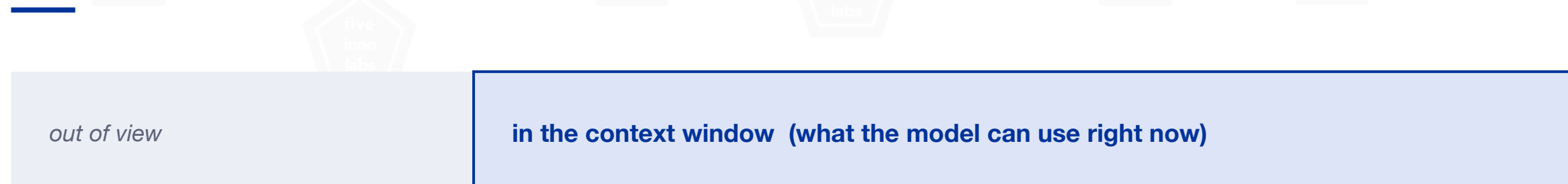
Every model has a token budget. Very long inputs get truncated.

Languages

Some languages use more tokens per word, so they cost more.

The model only sees what fits.

Your prompt, your files and every reply share one window. Past the limit, the earliest text falls out of view.



- **Long chats forget the start**

If a conversation runs long, restate the key facts near the end.

- **Big files can overflow**

A model with a small window cannot reason over a whole long document at once.

- **Put what matters last**

Recent context carries the most weight. Lead with the ask, not the backstory.

One dial: predictable or surprising.

Same prompt, two settings. Low temperature stays safe. High temperature roams.

TEMPERATURE 0 · most predictable

Prompt: "Give me a name for a coffee shop."

- The Daily Grind
- The Daily Grind
- The Daily Grind

Repeatable and on-task. Use for facts, code, extraction, anything you want the same way twice.

TEMPERATURE 1.2 · more random

Prompt: "Give me a name for a coffee shop."

- Cloud & Cardamom
- The Velvet Bean
- Loiter House Coffee

Varied and creative, but less reliable. Use for brainstorming, names, first drafts, ideas you will sift.

Most chat apps hide this dial. Knowing it exists explains why the same prompt can answer differently each time.

Made-up answers are a feature, not a glitch.

A model that always predicts the next likely word will sometimes predict a confident, plausible falsehood.

WATCH IT HAPPEN

You ask:

"What was the title of the 2019 paper on agentic compliance by author X?"

It answers:

"'Agentic Compliance Frameworks for Regulated AI', Journal of AI Policy, vol. 12."

Specific. Confident. And entirely invented. The paper, the journal and the volume do not exist.

Three signatures to watch for

- **Too specific**

Exact figures, names, dates or citations that would be hard to know.

- **Too confident**

No hedging at all on something genuinely uncertain or obscure.

- **Unverifiable**

You cannot find an independent source that confirms it.

You now know the mechanics.

The rest of the badge is about putting them to work.

- **Today** Spot one hallucination in the wild and name its signature.
- **This week** Notice tokens and context the next time a long chat drifts.
- **Next month** Try temperature 0 vs high on the same prompt and compare.
- **Beyond** Carry the model into every lesson. Mechanics first, then craft.

Next: The Evaluator's Mindset — use what you just learned to judge any AI output before you trust it.